Cosine Transform and Minority Graph Based Network for Imbalanced Image Classification

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Abstract
Although deep learning models show superior performance in image classification task but they usually fail in class imbalance situations where the model is biased toward the majority class during the training process. To address this issue in this work, it is suggested that relationships among minority classes are explored, and cross-correlations among these relationships with the given sample are extracted. To this end, a graph with learnable adjacency matrix is composed from centers of minority classes and processed through a light graph convolutional network. Then, the cross-attention mechanism is used to find the cross-correlations between the processed input and the minority graph features. All the processes are implemented in the compressed feature space obtained by the cosine transform, which significantly reduce data dimensionality and computational complexity. The minority graph features and cross-attention features are fused and injected to a multi-layer perceptron with two softmax activation functions where probabilities of all classes obtained by the initial softmax are integrated with the normalized cross-attention features through a residual connection and used for final classification through the second softmax function. The experimental results show superior performance of the proposed network for imbalanced data classification in terms of both classification accuracy and computational time.

Keywords: imbalanced image classification, graph convolutional network, cross-attention, discrete cosine transform.

1. Introduction

Deep learning models have widely used for different applications of image processing in recent years. But, because of reasons such as expensive manual annotation with costly energy, time and financial resources or presence of rare conditions or objects in some situations, we face with imbalanced datasets in applications such as medical image classification, anomaly detection and fraud detection [1]-[2].

Imbalanced image classification is a hot and challenging problem which two categories of data-level and algorithm-level methods have been suggested to deal with it. While in the first category, different re-sampling methods and synthesis processes have been used to balance the number of samples in different classes, in the second approaches, the classification algorithm is enhanced to handle the imbalanced datasets [3]-[4].

Various data generation methods have been introduced so far. The extreme machine learning autoencoder is used for sample generation in [5]. Auxiliary images are generated by transferring style of training samples in [6]. This generation method helps to balanced training of the model through style knowledge distillation. Some recent researches focus on automatically learning the augmentation policy. In [7], the sample-customized implicit semantic transformations are exploited as well as the adversarial learning to provide diversified data augmentation. An enhanced generative adversarial network (GAN) model is introduced in [8], which utilizes the isolation forest for identification of sparse region samples, and boundary sample detection method for detection of boundary samples within classes. Moreover, it uses the support vector data description for noise filtering of the generated samples.

Inner-class heterogeneity is usually ignored in imbalanced data classification methods. Samples of some majority classes are close to the those of minority ones which leads to misclassification. It is suggested in [9] that according to the inner-class heterogeneity, the majority class samples are divided into multiple fine-grained groups. Generation of augmented samples with sufficient diversity is struggled in traditional data augmentation methods. But usually, these techniques not only are not a robust solution but also lead to increasing the computational complexity and overfitting. GAN with learning the global distribution of data samples can generate more effective samples. However, conventional GANs balance just inter-class distributions while the intra-class model collapse may be occurred due to intra-class sparse samples. To deal with this difficulty, the cluster based local outlier factor method is suggested in [10] for identification of dense and sparse intra-class. Then, the GAN model is trained using these dense and sparse samples as conditions to appropriately focus on fitting sparse samples within the classes. Finally, the pure augmented samples are obtained by applying a one-class support vector machine (SVM) as a noise filter.

In addition to data-level methods, there are other approaches for imbalanced data classification which focus on improving the classification algorithm to avoid biasing to majority classes [11]-[12]. Some methods try to correct the loss function [13]-[14]. The cross-entropy ignores the output scores on incorrect classes. In [15], the complement cross-entropy is suggested and it is shown that considering the predicted probabilities on incorrect classes in the loss function can enhance the predication accuracy of the imbalanced image classification.

A convolutional neural network (CNN) model is suggested for imbalanced data classification in [16]. The noise is added to the feature space of CNN without changing ratio of minority and majority data. Noise addition in feature space of CNN leads to obtaining proper features during the training process. Moreover, a hybrid loss function of KL divergence and cross-entropy is suggested, which improves the minority class accuracy.

A hybrid imbalanced classification method is introduced in [17]. At first, a deep neural network is used to provide the embedding representation of data where the triplet loss is used to enhance differences between classes and similarities within classes. The oversampling in the embedding space is performed to alleviate effects of imbalanced data distribution; and a feature and center distance based oversampling method is also used to obtain more diverse samples. A cost-sensitive learning based fuzzy SVM is used as the final classifier to assign higher misclassification cost to samples of minority class.

Re-sampling and cost-sensitive learning methods need prior knowledge for training the model. To address this issue, dynamic curriculum learning is proposed in [18] to adaptively adjust the loss weight and sampling strategy in each batch where the curriculum learning is done in two levels: 1) sampling schedular that handles the data distribution not only from easy to hard but also from imbalance to balance, and 2) loss schedular that handles the learning importance between metric learning loss and classification.

To handle imbalance in labeled data, semi-supervised deep learning can be implemented to use the unlabeled data to help the model in learning from labeled samples. As a semi-supervised deep learning, an attention based label consistency model is introduced in [19]. The channel and sample attention exploit the relationships among different samples. An imbalanced semi-supervised learning with a graph based balanced classifier is introduced in [20]. To handle difficulty of label information propagation in tail classes, a graph based classifier head is attached to the representation layer to improve the pseudo-label propagation. Because the learning status of classes may vary, a flexible threshold adjustment is introduced in pseudo-labeling for selecting balanced samples participating in training. Moreover, a class-aware feature MixUp augmentation method is advised to deal with overfitting problem in tail classes. The semi-supervised learning algorithm is improved by adding an auxiliary balanced autoencoder branch in [21]. A class-aware reconstruction loss is advised to train the autoencoder to alleviate the confirmation bias through adaptive feature augmentation for various classes. Moreover, a graph based label propagation is adopted at the end of the autoencoder for more smoothing the output.

A transfer learning classifier is introduced for imbalanced data classification in [22]. It adjusts the number of samples with skewed distribution. Using an active sampling module, it expands the data samples through a data augmentation module and uses the DensNet model pre-trained on ImageNet dataset. Few-shot learning methods try to extract generalizable prior knowledge from the majority classes and transferring knowledge to the rare classes. However, these methods do not fully use the precise samples of minority classes and require the labeled base database. To deal with these problems for rare disease identification, a self-supervising contrastive loss based unsupervised representation learning is introduced in [23]. Moreover, the unsupervised representation learning is integrated with pseudo-label supervised classification; and assesses the intra-class diversity by utilizing the feature dispersion for dispersion-aware correction to decrease the inter-class performance imbalance.

Although various methods as represented in above have been suggested for imbalanced data classification but they mostly ignore the relationships among the minority classes. The proposed method in this work is based on this idea. If the relationships contained in the minority classes are explored, and the cross-correlations between the sample under test and the minority classes' relationships is mined, the network avoids to bias toward majority classes. The Cosine Transform and Minority Graph based Network (CTMGN) is introduced to address this problem. The main contributions of CTMGN are as follows:

1) The cosine transform is applied to the input images to convert them to the cosine feature space and reduce dimensionality. This dimensionality reduction leads to reduction of computational complexity and increasing the implementation speed in both training and prediction phases.
2) The label of each minority class is added to its features in cosine feature space to provide the feature matrix of minority classes.
3) To allow the network to learn the relationships among different minority classes, a learnable adjacency matrix is generated, which is learned during the training process.
4) A graph convolutional network is designed, which takes the obtained feature matrix and adjacency matrix as inputs and generates the minority graph features.
5) To find the cross-correlations and dependencies between the input sample and the minority classes, the cross-attention mechanism is applied to the minority graph features and the processed input sample.
6) Fusion of minority graph features and cross-attention features are used for data classification.
7) In addition to the highest probability obtained by the initial softmax, all other computed probabilities are also contributed in the final classification by the second softmax activation function.

The proposed method is described with details in section 2 and evaluated in section 3 with doing ablation study and comparison with related networks. Section 4 concludes the paper.

2. Proposed

The Cosine Transform and Minority Graph based Network (CTMGN) is introduced for imbalanced image multi-class classification in this section. The input RGB images are firstly compressed by using the cosine transform. To handle the imbalanced data and avoid biasing toward majority classes, the idea is to find relationships among the minority classes, and then, explore dependencies between the input sample with the minority classes' relationships. For arriving this aim, the graph convolutional network (GCN) [24]-[25] with learnable adjacency matrix is designed to extract the minority classes' relationships. Moreover, the cross-attention mechanism [26]-[27] is applied to explore correlations between the minority graph features with the input data.

The block diagram of the proposed framework is shown in Fig. 1. The RGB images are converted to the cosine transform, compressed and normalized. The reduced data is then used to generate the feature matrix of minority classes $(\boldsymbol{X})$. The learnable adjacency matrix $(\boldsymbol{A})$ is also provided by processing $\boldsymbol{X}$. After that, $\boldsymbol{X}$ and $\boldsymbol{A}$ are given to the GCN to find the relationships among the minority classes by extracting the graph features. Then, the cross-attention mechanism (CAM) is applied for exploring correlations and dependencies between the input data and graph features of minority classes. After that, the cross-attention (CA) features and the processed input data are normalized and fused together. The fused features are given to the output block where not only the maximum probability but also the probabilities of all classes are contributed in the classification process. Each part of the proposed method is explained with more details in the following.

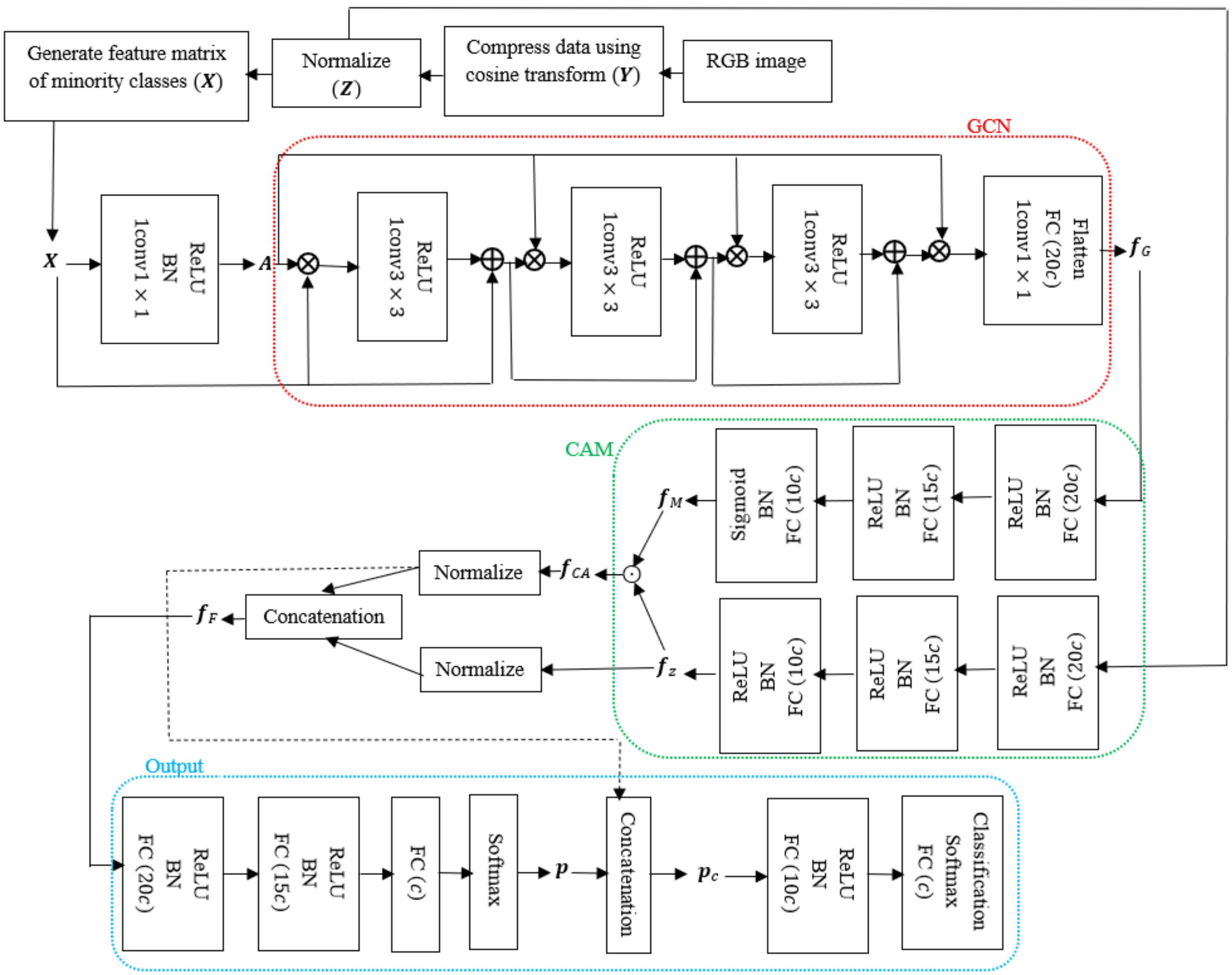

Fig. 1. Block diagram of the proposed framework.

2.1. Cosine transform for image compression

To compress each input RGB image with three channels, it is firstly converted to the gray scale image with one channel $\boldsymbol{I} \in \mathcal{R}^{a \times b}$ where a and b are the number of rows and columns, respectively. The discrete cosine transform (DCT) of the input image $\boldsymbol{I}$ with I_{ij} elements is computed by [28]-[29]:

$$B_{pq} = u_p u_q \sum_{i=0}^{a-1} \sum_{j=0}^{b-1} I_{ij} \cos\frac{\pi(2i+1)p}{2a} \cos\frac{\pi(2j+1)q}{2b}; 0 \le p \le a-1, 0 \le q \le b-1 \quad (1)$$

where B_{pq} is the element of pth row and qth column of the output coefficient matrix $\boldsymbol{B}$ and:

$$u_p = \begin{cases} \frac{1}{\sqrt{a}}, & p = 0 \\ \sqrt{\frac{2}{a}}, & 1 \le p \le a-1 \end{cases} \quad (2)$$

and

$$u_q = \begin{cases} \frac{1}{\sqrt{b}}, & q = 0 \\ \sqrt{\frac{2}{b}}, & 1 \le q \le b-1 \end{cases} \tag{3}$$

In the cosine coefficient matrix $\boldsymbol{B}$, the coefficients situated close to the upper-left corner are related to the lower frequencies, and the others correspond to higher frequencies, which can be discarded for image compression. To compress the input image $\boldsymbol{I}$, $\frac{1}{8}$ of the cosine coefficients in $\boldsymbol{B}$ are remained here and the other ones are discarded. The remained elements of the cosine transform containing the low frequency information constitute the compressed feature vector $\boldsymbol{y} \in \mathcal{R}^{d\times 1}$ with $d = \frac{(a\times b)}{8}$ features. So, the RGB image with $a \times b \times 3$ pixels is reduced to a vector consisting of d features. For all N image samples, the image compression is done and the reduced cosine feature space matrix $\boldsymbol{Y} \in \mathcal{R}^{d\times N}$ is obtained. The features are normalized along N samples as follows:

$$\boldsymbol{z}_i = \frac{y_i - \mu_i}{\sigma_i}; i = 1, \dots, d \tag{4}$$

where

$$\mu_i = \frac{1}{N}\sum_{j=1}^{N} y_{ij} \tag{5}$$

$$\sigma_i^2 = \frac{1}{N}\sum_{j=1}^{N}\left(y_{ij} - \mu_i\right)^2 \tag{6}$$

where $\boldsymbol{y}_i; i = 1, \dots, d$ is ith row of $\boldsymbol{Y}$ associated with ith feature, and $\boldsymbol{z}_i; i = 1, \dots, d$ is its normalized version, which composes ith row of the normalized compressed data matrix as $\boldsymbol{Z} \in \mathcal{R}^{d\times N}$.

2.2. Minority graph

To explore the complex relationships among data samples of the minority classes, the minority graph is constructed and processed by a GCN. The designed GCN has two inputs: the feature matrix ($\boldsymbol{X} \in \mathcal{R}^{m\times d}$), and the learnable adjacency matrix ($\boldsymbol{A} \in \mathcal{R}^{m\times m}$) where m is the number of graph nodes and d is the number of features. In conventional GCNs, the adjacency matrix is computed by applying a similarity measure to find the relationships among each pair of nodes. But, considering this approach may not explore the hidden nonlinear and complex relationships between the nodes. The number of nodes (m) is considered as the number of minority classes here. The centers of m minority classes obtained from the compressed data ($\boldsymbol{Z}$) are used to compose the feature matrix $\boldsymbol{X} = [\boldsymbol{x}_1\ \boldsymbol{x}_2 \dots \boldsymbol{x}_m]^T$ where $\boldsymbol{x}_i \in \mathcal{R}^{d\times 1}; i = 1, \dots, m$ is center of ith minority class. The feature matrix $\boldsymbol{X}$ is normalized along d features to allocate in range of [0,1].

To include the minority class labels to the feature matrix of the minority classes, elements of the matrix $\boldsymbol{X}$ is reformed as follows:

$$x_{ij} = \log\left[x_{ij} + i\right]; i = 1, \dots, m; j = 1, \dots, d \tag{7}$$

where x_{ij} is element of $\boldsymbol{X}$ in ith row and jth column. i that is the corresponding label associated with features of ith minority class is added to the normalized feature x_{ij}.

To provide the learnable adjacency matrix, the feature matrix $\boldsymbol{X}$ is passed from a convolutional filter with kernel size of 1×1 denoted by $1\text{conv}1 \times 1$ followed by batch normalization (BN) and ReLU as the nonlinear activation function. So, the learnable adjacency matrix is obtained by:

$$\boldsymbol{A} = g\left[\mathcal{N}\left(1\text{conv}1 \times 1(\boldsymbol{X})\right)\right] \tag{8}$$

where $\mathcal{N}(\cdot)$ is the BN and $g[\cdot]$ is the ReLU function. After generation of the adjacency matrix, the feature matrix $\boldsymbol{X}$ and the adjacency matrix $\boldsymbol{A}$ are given to the designed GCN where features of $(l+1)$layer are extracted by:

$$\boldsymbol{H}_{l+1} = g[1\text{conv}3 \times 3(\boldsymbol{A}\boldsymbol{H}_l)] + \boldsymbol{H}_l \tag{9}$$

where $1\text{conv}3 \times 3$ means a convolutional filter with kernel size of 3×3, and $\boldsymbol{H}_l$ is features of lth layer and $\boldsymbol{H}_1 = \boldsymbol{X}$. The designed GCN has three layers from the above type. After extraction of $\boldsymbol{H}_3$, it is multiplied in $\boldsymbol{A}$ and passed from a $1\text{conv}1 \times 1$, a fully connected (FC) layer with $20c$ units where c is the number of classes and flatten layer to generate the graph features $\boldsymbol{f}_G$.

2.3. Cross-attention mechanism

For exploring relationships between the input image with different minority classes to avoid biasing the network training toward the majority classes, the cross-attention mechanism (CAM) is applied among the features of input image with features of the minority graph. From one hand, the compressed input image obtained by the cosine transform is passed from a multi-layer perceptron (MLP) consisting of three FC layers where each FC layer is followed by a BN and ReLU layer. The FC layers contain $20c$, $15c$ and $10c$ units, respectively, where c denotes the number of classes as before. This MLP network is used for feature extraction from the input feature vector $\boldsymbol{z}$. The extracted features are denoted by $\boldsymbol{f}_z$. From the other hand, the graph features $\boldsymbol{f}_G$ extracted from the previous stage are also passed from a similar MLP network containing 3 FC layers with $20c$, $15c$ and $10c$ units, respectively where two first FC layers are followed by the BN and ReLU layers, and the last FC layer is followed by BN and sigmoid activation function to generate weights in interval of [0,1].

The weights obtained from the graph features, which show importance of the minority classes, are denoted by $\boldsymbol{f}_M$. The cross-correlation among the input features $\boldsymbol{f}_z$ and the features of the minority classes, $\boldsymbol{f}_M$, is obtained by:

$$\boldsymbol{f}_{CA} = \boldsymbol{f}_z \odot \boldsymbol{f}_M \tag{10}$$

where $\boldsymbol{f}_{CA}$ indicates the cross-attention (CA) features, and $\odot$ represents the elementwise multiplication.

2.4. Feature fusion and output

The CA features $\boldsymbol{f}_{CA}$ and the processed input features $\boldsymbol{f}_z$ are normalized through passing the BN layer, and then, concatenated together to provide the fused features $\boldsymbol{f}_F$ as follows:

$$\boldsymbol{f}_F = \text{concat}\,[\mathcal{N}(\boldsymbol{f}_{CA}), \mathcal{N}(\boldsymbol{f}_z)] \tag{11}$$

where concat $[\cdot,\cdot]$ indicates the concatenation operation. The fused features are given to the output network, which is a MLP with residual connection. The output network starts with two FC layers with $20c$ and $15c$ units each one followed by BN and ReLU. The network is continued with a FC with c units and softmax activation function to generate probabilities of c classes. Instead of using just the highest probability in output of the softmax function, which is likely associated with the true class, the probabilities of all c classes are used to provide the final class label. To this end, the probabilities $\boldsymbol{p}$ obtained in output of the initial softmax activation function are concatenated with the normalized cross-attention features to provide complete distribution features as follows:

$$\boldsymbol{p}_c = \text{concat}\,[\mathcal{N}(\boldsymbol{f}_{CA}), \boldsymbol{p}] \tag{12}$$

The complete distribution features $\boldsymbol{p}_c$ are passed from a FC with $10c$ units, BN and ReLU, and then, passed from a FC with c units, softmax and classification layer to obtain the output.

3. Experiments

3.1. Datasets

Two known image datasets are used to evaluate performance of the proposed method in this work. The Street View House Numbers (SVHN) contains house numbers in Google Street View images [30]. The cropped images with 32×32 pixels are used. This dataset has 73257 samples for training and 26032 ones for test. In this work, a small subset of SVHN containing 25% of samples of each class are used. The CIFAR-10 dataset contains RGB images with 32×32 pixels consisting of 10 classes from everyday objects [31]. This dataset contains 5000 training samples per class totally 50000 training samples and 1000 ones per class for test, i.e., totally 10000 test samples.

To compose the imbalanced datasets, all samples of first five classes and a fraction of samples in last five classes in SVHN; and all samples of last five classes and a fraction of first five classes in CIFAR-10 are considered in both training and test sets and the remained samples are discarded where the imbalance ratio ρ is defined by:

$$\rho = \frac{\max_{i=1,\dots,c} n_i}{\min_{i=1,\dots,c} n_i} \tag{13}$$

where n_i denotes the number of samples in ith class. The experiments are done with $\rho = 14.87$ for SVHN dataset, and implemented with $\rho = 10$ for CIFAR-10 dataset.

For SVHN dataset, the data augmentation is applied for minority classes where noisy images with Gaussian distribution, zero mean and standard deviation in interval of [0.0005,0.3] are added to the original data. All experiments are done in MATLAB R2022b with a 13th Gen Intel Core i5 processor and 32 GB RAM.

3.2. Ablation study

To assess impact of the main parts of the proposed network, the following cases are compared together:

- Cosine Transform based Network (CTN): in this case, the input sample, which is the compressed vector of the input image obtained by the cosine transform, is processed by using a MLP network consisting of three FC layers each one followed by a BN and ReLU with $20c$, $15c$ and $10c$ units, respectively. The extracted features are normalized, and then, given to the output network. Similar to Fig. 1, the output network starts with two FC layers followed by BN and ReLU with $20c$ and $15c$ units. Then, a FC with c units and initial softmax activation function is applied to compute probabilities of c classes. Then, the obtained probabilities are concatenated with normalized input features processed by the MLP through a residual connection. The network is ended with a FC with $10c$ units, BN, ReLU, a FC with c units, softmax and classification layer.
- Minority Graph based Network (MGN): in this case, the minority feature matrix $\boldsymbol{X}$ is generated and used to provide the learnable adjacency matrix $\boldsymbol{A}$. Then, $\boldsymbol{X}$ and $\boldsymbol{A}$ are given to GCN similar to Fig. 1 where the considered GCN is ended with $1\text{conv}1 \times 1$, a FC with $20c$ units and flatten layer to provide the graph features $\boldsymbol{f}_G$. The graph features are passed from a MLP with three FC layers where the last FC layer is followed by BN and sigmoid activation function to generate the graph weights $\boldsymbol{f}_M$ in [0,1]. Moreover, the compressed input features are processed by passing a MLP to provide $\boldsymbol{f}_z$ features. The $\boldsymbol{f}_z$ and $\boldsymbol{f}_M$ features are multiplied elementwise to generate the cross-attention features $\boldsymbol{f}_{CA}$. The extracted CA features are given to the output network where the normalized $\boldsymbol{f}_{CA}$ features are concatenated with probabilities in output of the initial softmax activation function through a residual connection.
- Cosine Transform and Minority Graph based Network (CTMGN): this is the complete version of the proposed network, which is shown in Fig. 1.

The ablation study results obtained for SVHN dataset are reported in Table 1. The Adam optimizer with mini-batch size of 128, learning rate of 0.001 and 25 epochs is used to train the network. As seen, the CTMGN method provides the highest accuracy in terms of all evaluation measures except the average precision. According to precision, MGN ranks first and with a bit difference, CTMGN ranks second. Moreover, MGN is preferred than CTN in terms of all measures. It means that the use of minority graph features and exploring their relationships with the input features can

Table 1. Ablation study results obtained for SVHN dataset.

	CTN	MGN	CTMGN
Overall Accuracy	78.08	78.50	**78.59**
Average Precision	63.77	**65.72**	65.46
Average Recall	65.61	66.60	**68.25**
Average Selectivity	77.98	78.41	**78.51**
Average Fmeasure	64.58	66.09	**66.65**
Average Gmean	71.00	71.78	**72.88**
Training time (s)	**189.98**	448.08	487.00
Test time (s)	**1.73**	3.27	3.22
No. of parameters	**135.4k**	342k	362.2k

Table 2. Ablation study results obtained for CIFAR-10 dataset.

	CTN	MGN	CTMGN
Overall Accuracy	52.91	53.82	**54.35**
Average Precision	33.95	34.93	**35.63**
Average Recall	34.68	35.54	**36.10**
Average Selectivity	52.83	53.74	**54.27**
Average Fmeasure	33.98	35.14	**35.79**
Average Gmean	39.62	40.76	**41.40**
Training time (s)	**219.78**	501.06	521.70
Test time (s)	**0.52**	1.01	1.08
No. of parameters	**135.4k**	342k	362.2k

positively enhance the classification performance. The training time, test (prediction) time and the number of learnable parameters are also represented in the last three rows. It can be found that the lightest network with the lowest running time is CTN. CTMGN has higher training time and more number of parameters than MGN but its prediction time is close to that of MGN.

The obtained classification results in ablation study for CIFAR-10 dataset are represented in Table 2. The Adam optimizer with learning rate of 0.01, mini-batch size of 128, and 30 epochs are used to train the network. As seen, CTMGN, MGN and CTN rank first to third, respectively, in terms of all measures of overall accuracy, average precision, recall, selectivity, Fmeasure and Gmean. The fastest method with the lowest number of parameters is CTN. MGN and CTMGN have higher running time in both training and prediction phases. However, they can be run in about 1 second in the test stage.

3.3. Comparison with other networks

The proposed method is compared with convolutional neural network (CNN), attention based CNN (ACNN) and graph convolutional network (GCN) in this section. All the competitors are trained by using the Adam optimizer, learning rate of 0.001, mini-batch size of 1024 and 25 epochs.

The implemented CNN takes the RGB image as the input, and consists of four convolutional layers with filter size of 3×3 with 8, 16, 32 and 64 filters, respectively where each convolutional layer is followed by BN for normalization. GeLU as the nonlinear activation function, and dropout layer with dropping probability of 0.5 to deal with overfitting problem. The network is ended with a FC with c units, softmax and classification layer.

The ACNN uses the fused features of convolution kernels and attention mechanism. The convolutional features are generated by a CNN similar to what described above. For implementation of the self-attention mechanism, the key (K), query (Q) and value (V) components are generated by applying $32\text{conv}3 \times 3$ to the input RGB images. Then, the attention value is obtained by the softmax $\left(\frac{K \odot Q}{\sqrt{s}}\right) \odot V$ where $\odot$ is the elementwise multiplication and s is the feature dimensionality. Output of CNN and attention networks are individually passed from a FC with $4c$ units followed by BN and then, added together to provide the fused features. Then, the fused features are given to a FC with c units ended with softmax and classification layers.

Table 3. Comparison of different methods for SVHN dataset.

	CNN	ACNN	GCN	CTMGN
Overall Accuracy	63.43	63.47	75.66	**78.59**
Average Precision	48.35	45.02	61.73	**65.46**
Average Recall	51.18	45.35	64.74	**68.25**
Average Selectivity	63.31	63.29	75.57	**78.51**
Average Fmeasure	47.31	43.06	62.65	**66.65**
Average Gmean	55.83	50.66	69.48	**72.88**
Training time (s)	865.54	1118.56	861.80	**487.00**
Test time (s)	34.70	29.53	9.92	**3.22**
No. of parameters	680.1k	3.9M	6.1M	**362.2k**

Table 4. McNemars test results of different methods for SVHN dataset.

	CNN	ACNN	GCN	CTMGN
CNN	0	-0.11	-30.57	-37.83
ACNN	0.11	0	-31.02	-38.24
GCN	30.57	31.02	0	-9.35
CTMGN	37.83	38.24	9.35	0

In the GCN method, a graph is constructed from the input images where the means of classes compose c nodes with $d = a \times b \times 3$ features where $a \times b$ is the spatial size of the input image and 3 corresponds to RGB channels. Associated with each input image, a graph with feature matrix of $\boldsymbol{X} \in \mathcal{R}^{c \times d}$ is composed where $\boldsymbol{x}_i = \frac{\boldsymbol{\mu}_i \odot \boldsymbol{y}}{\|\boldsymbol{y}\|}$ and $\boldsymbol{y} \in \mathcal{R}^{d \times 1}$ is the input feature vector, i.e., the reshaped version of the input image, $\boldsymbol{\mu}_i \in \mathcal{R}^{d \times 1}; i = 1, \dots, c$ is the mean of ith class, and $\boldsymbol{x}_i \in \mathcal{R}^{d \times 1}$ is ith row of the feature matrix $\boldsymbol{X}$. The adjacency matrix $\boldsymbol{A} \in \mathcal{R}^{c \times c}$ is also computed from each pair of the feature matrix rows as follows:

$$A_{ij} = \frac{(x_i - x_j)^T (x_i - x_j)}{\|x_i\| \|x_j\|} \tag{14}$$

where A_{ij} is element of $\boldsymbol{A}$ in ith row and jth column. $\boldsymbol{X}$ and $\boldsymbol{A}$ are given to a GCN with three layers where the extracted graph features are given to a MLP for more feature extraction. The MLP has three FC layers with $20c$, $15c$ and $10c$ units, respectively where each FC is followed by BN and ReLU, and the network is ended with a FC with c units, softmax and classification layer.

Comparison of different methods for SVHN dataset is represented in Table 3. The proposed CTMGN ranks first in terms of all evaluation measures. To assess that difference of methods is statistically significant or not, the McNemars test is done. According to the McNemars test results in Table 4, preference of CTMGN compared to all competitors is statistically significant with Z score of Z=37.83 corresponding to CNN, Z=38.24 associated with ACNN and Z=9.35 related to GCN. The second rank with significant difference compared to CNN and ACNN is GCN.

In terms of average Gmean, Fmeasure, recall, and precision, CNN is preferred than ACNN, and in terms of average selectivity and overall accuracy, their performances are close together, and according to the Z score where |Z|=0.11<1.96 their difference is not significant. The proposed CTMGN method, which uses the compressed images obtained by the cosine transform as input of the network is the lightest network with the lowest number of parameters. It has much less computation time in both training and prediction phases that can be implemented in about 3 seconds in the test stage. After CTMGN, the lightest network is CNN. However, it needs the highest test time. GCN and ACNN with 6.1M and 3.9M parameters have the highest number of parameters, respectively while the highest training time belongs to ACNN.

The classification results and the obtained Z scores for CIFAR-10 dataset are reported in Tables 5 and 6, respectively. The highest accuracy in terms of different measures is obtained by CTMGN where its difference compared to CNN and ACNN with Z=20.99 and Z=12.41 is significant, and its difference compared to GCN with Z=1.51<1.96 is not significant. The second rank in terms of all measures except average recall is GCN where the

Table 5. Comparison of different methods for CIFAR-10 dataset.

	CNN	ACNN	GCN	CTMGN
Overall Accuracy	35.80	44.02	53.13	**54.35**
Average Precision	31.85	29.99	33.96	**35.63**
Average Recall	34.27	27.18	34.17	**36.10**
Average Selectivity	35.79	43.95	53.05	**54.27**
Average Fmeasure	27.54	27.06	34.02	**35.79**
Average Gmean	34.08	30.66	38.97	**41.40**
Training time (s)	655.05	914.27	773.97	**521.70**
Test time (s)	10.08	11.57	2.92	**1.08**
No. of parameters	680.1k	3.9M	6.1M	**362.2k**

Table 6. McNemars test results of different methods for CIFAR-10 dataset.

	CNN	ACNN	GCN	CTMGN
CNN	0	-10.03	-19.52	-20.99
ACNN	10.03	0	-11.07	-12.41
GCN	19.52	11.07	0	-1.51
CTMGN	20.99	12.41	1.51	0

Table 7. Performance of CTMGN in different imbalanced ratios for SVHN dataset.

	$\rho = 5.94$	$\rho = 14.87$	$\rho = 29.62$	$\rho = 59.74$	$\rho = 88.85$	$\rho = 119.48$	$\rho = 150.65$
Overall Accuracy	76.00	78.59	78.97	79.55	82.27	78.63	79.72
Average Precision	69.36	65.46	59.08	52.38	50.89	46.46	46.39
Average Recall	71.17	68.25	58.29	52.18	49.48	46.86	45.46
Average Selectivity	75.92	78.51	78.86	79.46	82.19	78.56	79.65
Average Fmeasure	70.12	66.65	58.57	52.13	50.09	46.47	45.69
Average Gmean	73.33	72.88	66.36	61.74	59.54	56.33	54.30

second rank according to recall is related to CNN. In terms of average Gmean, Fmeasure, recall and precision, CNN is better than ACNN, and in terms of overall accuracy and average selectivity, ACNN is preferred than CNN where this difference with Z=10.03>1.96 is statistically significant. CTMGN has the lowest number of learnable parameters, which runs in about 1 second in the prediction phase. GCN although has the highest number of parameters is the fastest method after CTMGN in the test stage, which can be run in 2.92~3 seconds in the prediction phase. Moreover, although CNN is relatively a light network compared to ACNN and GCN but it has relatively a high prediction time.

3.4. Assessment in different imbalanced ratios

To assess performance of the proposed CTMGN method in different imbalanced ratios, its performance is obtained in different ρ values, and the achieved results for SVHN dataset are reported in Table 7. As seen, with increasing the imbalanced ratio ρ, average precision, recall, Fmeasure and Gmean values are decreased. With the imbalanced ratio of $\rho = 88.85$, the highest overall accuracy is obtained. It shows that overall accuracy cannot be an appropriate measure for classification assessment in imbalanced data.

4. Conclusion

An imbalanced image classification method is proposed in this work, which tries to explore the relationships among the minority classes through composing a graph with learnable adjacency matrix in the cosine transform feature space. The ablation study is done that shows positive effect of the minority graph features in improving the classification results. Moreover, comparisons with other networks such as CNN, attention based CNN and graph convolutional network show preference of the proposed method in terms of different evaluation measures. The proposed network has a lightweight model, which runs relatively fast specially in the prediction phase. Improvement of the graph design from both feature matrix and adjacency matrix point of views may enhance the classification results, which can be studied in future works.

Data Availability

The datasets used for the experiments are benchmark datasets.
The SVHN dataset is available online in http://ufldl.stanford.edu/housenumbers/.
The CIFAR-10 dataset is available online in https://www.cs.toronto.edu/~kriz/cifar.html.

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